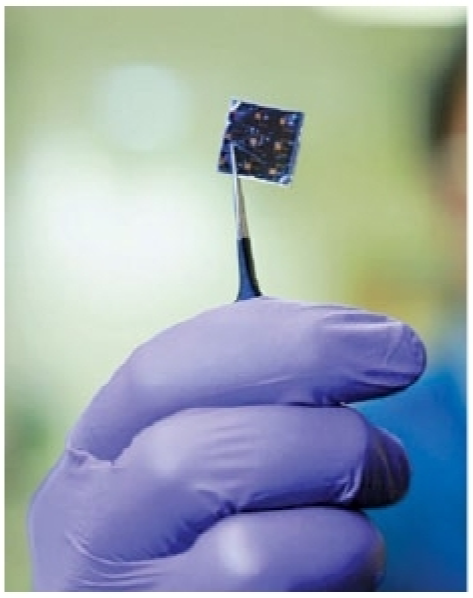
Quality and speed of data transmission are achieved based on the superior constrained shortest path finder algorithm, made by the scientists. Thus, data transmission speed can be increased up to 50 per cent.

Co-author Andrey Sukhov explains with an example, "The mechanism can be used by scientists who conduct experiments at Large Hadron Collider at CERN. They calculate tasks in laboratories scattered all over the world, and make inquiries to computer centres of CERN. They also exchange both textual information and high-resolution streaming video online. The technology helps them with this." The algorithm can also be used by specialists involved in International Thermonuclear Experimental Reactor, which is being constructed at CEA research centre in France.

Dual-layer solar cell for efficiently generating power

Materials scientists from UCLA Samueli School of Engineering have developed a highly efficient thin-film solar cell that generates more energy from sunlight than typical solar panels, thanks to its double-layer design.



The solar cell developed by UCLA Engineering researchers converts 22.4 per cent of incoming energy from the Sun—a record in power conversion efficiency for a perovskite-CIGS tandem solar cell

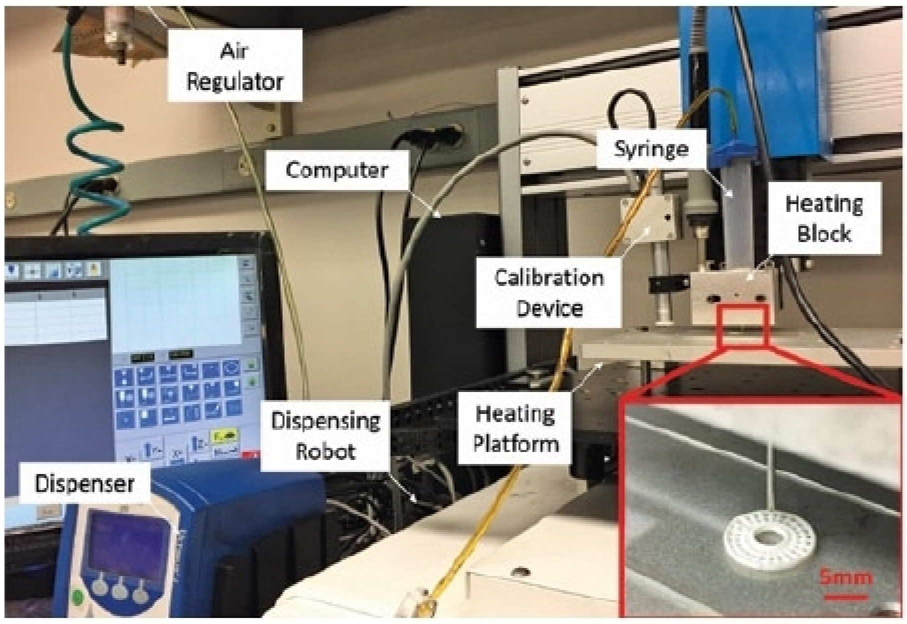
The device is made by spraying a thin layer of perovskite—an inexpensive compound of lead and iodine that is efficient at capturing energy from sunlight—onto a commercially available solar cell. The solar cell that forms the bottom layer of the device is made of a compound of copper, indium, gallium and selenide, or CIGS.

Yang Yang, UCLA's Carol and Lawrence E. Tannas Jr. Professor of Materials Science, says, "With our tandem solar cell design, we are drawing energy from two distinct parts of the solar spectrum over the same device area. This increases the amount of energy generated from sunlight compared to the CIGS layer alone."

The new cell converts 22.4 per cent of incoming energy from the Sun. The performance was confirmed in independent tests at U.S. Department of Energy's National Renewable Energy Laboratory. The previous record, set in 2015 by a group at IBM's Thomas J. Watson Research Center, was 10.9 per cent.

3D-printed electrolyte for lithium-ion batteries

Researchers at University of Illinois at Chicago (UIC) College of Engineering have printed a stable, yet flexible, solid-state electrolyte using an elevated-temperature extrusion printing technique. Reza Shahbazian-Yassar, associate professor, and Yayue Pan, assistant professor of mechanical and industrial engineering in UIC College of Engineering, along with colleagues have designed a unique 3D printer that works at an elevated temperature. It extrudes materials at around 120°C compared to traditional extrusion 3D printing, which takes place at room temperature.



The 3D printing setup

The solid-state electrolyte ink is made of a polymer base containing titanium oxide particles that allow it to be flexible as well as functional. It can be directly deposited into the battery as it is being printed.

According to Meng Cheng, graduate student in UIC College of Engineering and lead author of the study, "High temperature prevents post-production shrinkage. Our technique dramatically improves efficiency of the preparation of the electrolyte and its incorporation into the battery."

Mass producing these batteries is a labourious and expensive process. 3D printing these makes the production and customisation quick and inexpensive, because all parts are printed together at once. In lithium-ion batteries, where electrodes have been 3D-printed, the conventionally produced electrolyte is added later in a separate step.

Organising quantum dots may improve consumer electronics

Prof. Michael Bockstaller and his team at Carnegie Mellon University Materials Science and Engineering are working on ways for quantum dots, or nanoparticles created from a specific semiconductor, to self-assemble into organised patterns. The research may help improve the quality of image displays for electronic devices such as televisions and tablets.

When the dots are surrounded by particles of the same colour, it minimises heat absorption and increases efficiency. The team has already proved that it can control these nanoparticles by attaching a polymer chain to these. Now, it plans to study exactly what about the polymer causes the particles to move.

According to Prof. Bockstaller, if you take a set of marbles